Extra exercises of Lecture 6

EX 1

Consider a government bond that pays 9.8% as interest rate and a 11.5% municipal bond. Which is the marginal tax rate that lets you indifferent between the two?

a) 15.2%
b) 17.9%
c) 12.5%
d) none of the solutions above

EX 2

A corporation sells today 10,000 coupon bonds, with maturity of 2 years at the current market rate (5.5%) and a face value of $1,000. The bonds have a call provision at par. At the beginning of the first year, the market interest rate is 5%. What is the profit if the corporation finds convenient to buy back the bonds?

a) 0 because it is not convenient to opt the buy back
b) $ 295,000.0
c) $ 47,619.0
d) none of the others

EX 3

Consider a three-year coupon bond with semi-annual coupon payments, with a face value of $1,500, a coupon rate of 12% and that the interest rate at which investors discount cash flows on similar bonds is 10%. What is the interest rate of a zero-coupon bond with the same maturity and face value of $1,800 that allows to sell it at the same price of the other bond today?

a) 11.2%
b) 4.5%
c) 2.1%
d) 5.4%

EX 4

Which of the following are true about bonds and money markets?

I. trade in both bonds and money markets instruments mainly takes place in over-the-counter exchanges
II. trade in bonds mainly takes place in organized exchanges
III. trade in money markets instruments mostly in over-the-counter exchanges
IV. trade in both bonds and money markets instruments mainly takes place in organized exchanges

a) II and III
b) I and III
c) II and IV
d) none of the others above